OMAR AHMED

Full-Stack Software Engineer | Node.js, React & Angular

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PROFILE

Software engineer with **3 years of experience** building frontend and backend applications across education, booking, nutrition, e-commerce and SaaS. Experienced in modular architecture, PostgreSQL performance, messaging and real-time systems, using React and Angular alongside Node.js and TypeScript.

TECHNICAL SKILLS

Backend: Node.js, NestJS, Express.js, TypeScript, JavaScript, REST, GraphQL
Architecture & messaging: Hexagonal architecture, CQRS, DDD, RabbitMQ, WebSockets
Data: PostgreSQL, TypeORM, MongoDB, Redis, MySQL, SQL Server, Kysely
Frontend & delivery: React, Angular, Next.js, Docker, AWS, Git, Jest, Testcontainers, k6

EXPERIENCE

Raise Right | Full-Stack Developer

Jun 2025 - Present | Full-time | Cairo, on-site

- Developed frontend and backend features for **EduMarket, partner service management, a partner B2B marketplace, Roli transportation, central Admin and Crew operations**. Roli connects drivers, supervisors and parents, with partner-side transportation management.
- Contributed to EduMarket's **58-module NestJS backend**. In load tests with **300 concurrent users**, raised throughput **6.8x** (4.85 to 32.8 req/s), cut p95 latency from **60s to 9.9s**, and reduced HTTP failures from **79.5% to approximately 0%**.
- Diagnosed bottlenecks using **pg_stat_statements**; used targeted indexes, DataLoader batching and tag-based Redis caching to reduce top query times from **6-15 seconds to sub-millisecond**, with a measured **99.9% cache-hit rate**.
- Restored enrollment and wallet flows from **0% to 100% success under load**; integrated Kashier and Paymob behind circuit breakers. Removed **41 circular module dependencies** and introduced dependency checks before commits and pushes.
- Supported reliability with **3,000+ tests across 428 Jest/Testcontainers suites** and **23 k6 scenarios**, including a 20-to-200 RPS spike and a 45-minute soak at 15 RPS. The **1,000,000-user dataset was seeded test data**. Used Sentry, Prometheus and Loki for observability.

Apes Smart Solutions | Full-Stack Developer

Feb 2025 - Dec 2025 | Part-time | Remote

- Delivered full-stack features across **3 product areas**: nutrition applications, Spark activity bookings, and Views property/hospitality workflows.
- Applied **hexagonal architecture** and RabbitMQ messaging; used **TypeORM with PostgreSQL** for relational data and **MongoDB for audit records**.
- Built services handling **2,000+ requests/second at 99.9% uptime**; used RabbitMQ to reduce synchronous blocking by **approximately 70%** and supported **1,000+ daily bookings**.
- Improved logging and tracing to reduce incident debugging time by **60%**.

Sarri Technology | Full-Stack Developer (MERN)

Oct 2023 - May 2025 | Full-time | Cairo, on-site

- Developed frontend and backend features and modernized a **10,000-MAU** application, reducing critical bugs by **40%** and API latency by **45%**; built multi-tenant e-commerce, booking and HR SaaS workflows.
- Built **40+ REST APIs** serving **5M+ daily requests at 99.99% uptime**; rolled out **v2 and v3** with zero-downtime deployments.
- Built analytics dashboards processing **100M+ data points/month**; worked with an **8-person frontend team** to accelerate feature delivery by **25%**.
- Delivered reusable payment and notification components, team/payroll workflows, WebSockets updates and Google Maps integrations.

CustEx | Full-Stack Developer

Aug 2023 - Nov 2023 | Part-time | Remote

- Built frontend and backend workflows for real-time interactions and voting, handling **15,000-20,000 messages/day** with **under 150ms delivery**; reduced connection latency by **25%** with **99.7% connection success**.

SELECTED PROJECTS

EduMarket | NestJS, TypeScript, PostgreSQL, Redis, GraphQL, Angular
Spark | NestJS, Node.js, MongoDB, Socket.io, React, Next.js
Views | NestJS, Node.js, Express, MongoDB, React
Jam3ty | NestJS, Node.js, TypeScript, MongoDB, Angular
Additional technologies: Python, FastAPI, Mosquitto, WebRTC, MQTT, AMQP, gRPC, Protocol Buffers; AWS S3, EC2, RDS, ECS, Lambda, Amplify and SQS. Agile development practices.

EDUCATION

Fayoum University | B.Sc. Computer Science, Artificial Intelligence | 2019-2023